Applied Information Extraction in Python

Today's goal • 2/3 items

Module 1
Information Extraction

Welcome to the Course
- Welcome to Information Extraction — Video • 6 min
- MADSwPy Certificate Roadmap — Reading • 10 min
- Course Syllabus — Reading • 10 min
- Introduction to Jupyter Notebook — Reading • 10 min
- Meet Other Learners — Discussion Prompt • 10 min
- Help Us Learn About You — Reading • 10 min

Introduction to Information Extraction
- What is Information Extraction? — Video • 8 min
- Information Extraction in Different Domains — Video • 6 min
- Knowledge Check: Introduction to Information Extraction — Practice Assignment • 10 min

Rule-Based Approaches to Information Extraction
- Extracting Formatted Information — Video • 7 min
- Regular Expressions in Detail — Reading • 20 min
- Lookup Based Extraction — Video • 4 min
- Knowledge Check: Rule-Based Approaches to Information Extraction — Practice Assignment • 10 min

How to Set Up Rule-Based Approaches in Python
- Demo: Using Regular Expressions & Examining Output — Video • 4 min
- Jupyter Notebook Practice on Basic NLP and Rule-Based Extraction — Lab • 1h

Module Wrap-Up & Assignments
- Assignment 1 Introduction: Formatting & Normalizing Data with Regular Expressions — Video • 4 min
- Build an Information Extraction Pipeline for Template/List-Based Fields — Programming Assignment • 3h
- Module 1 Assignment — Graded Assignment • 30 min

Module 2
Named Entity Recognition (NER)

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

Course Syllabus

0:29 / 12:28 2x

Welcome!

Welcome to Applied Information Extraction in Python, taught by V. G. Vinod Vydiswaran!

In this course, you will explore a range of techniques and tools for extracting meaningful information from text. We will begin by working with well-structured, easily recognizable patterns, such as those in dates or phone numbers, using tools such as regular expressions, lists, and gazetteers. As we progress, you will learn how to extract more complex information through Named Entity Recognition (NER), applying machine learning techniques to identify entities such as people, organizations, and locations based on their linguistic context. Building on this, we will dive deeper into neural network models that leverage non-linear transformations to detect subtle patterns and relationships within text. Finally, we will examine cutting-edge Transformer-based models, including Large Language Models like GPT, which have dramatically advanced the capabilities of information extraction at scale. Throughout the course, you will gain practical experience applying these approaches across diverse domains such as healthcare, literature, and postal systems.

This is the fourth course in the **More Applied Data Science with Python** specialization from the University of Michigan. We recommend that you take this course after completing Applied Unsupervised Learning in Python. We also suggest that before taking this course that you complete the **Applied Text Mining in Python** course from the Applied Data Science with Python (ADSwPy) series.

Participation Strategies

Engaged learning looks different for everybody. In this course, we hope you will define your own measures of success and engage with the material in a way that best suits your needs.We recognize and celebrate the diverse ways learners engage in courses. As you go through this course, we hope you will reflect on your unique skills, needs, and aspirations, and engage in the course material in a way that aligns with your own goals. While the course provides time estimates for completion, you should feel empowered to engage in the material in whatever ways make sense to you.

Course Structure & Grading Policy

The course is structured into four modules. In each module, we have included lecture videos, ungraded knowledge check assignments, auto-graded assignments, and programming assignments.

Module 1: Information Extraction

This module introduces information extraction, covering key tasks and approaches for extracting relevant information from text. You will explore pattern-based and list-based methods to identify and extract information from text data, applying these techniques across diverse domains. You will also develop an end-to-end NLP pipeline to extract named entities from free text using terminology resources.

- Graded Assessment
  - Module 1 Assignment (5 % of total grade)
  - Programming Assignment: Build an Information Extraction Pipeline for Template/List-Based Fields (20 % of total grade)

Module 2: Named Entity Recognition (NER)

In Module 2, you'll dive into the world of named entity recognition (NER). You'll learn to define and identify named entities, and understand how to tackle related tasks by framing them as NER challenges. We'll explore how to use resources like standardized terminology and named gazettes to enhance NER. You'll also gain hands-on experience by training a machine learning model for sequence classification using an annotated text dataset. Finally, we'll discuss the pros and cons of different Markov models for NER, equipping you with the insights needed for practical applications.

- Graded Assessment
  - Module 2 Assignment (5 % of total grade)
  - Assignment: Build an Information Extraction Pipeline for CRF Based Extraction (20 % of total grade)

Module 3: Neural Network Models

In Module 3, focused on neural network models, you will explore the differences between training deep learning models and traditional machine learning models. You'll learn how to model and train a neural network-based classifier, as well as formulate text as features for NER model training. We will discuss the pros and cons of deep learning approaches. You'll design a neural network model to identify concepts from free text and apply a trained deep learning model to solve NER tasks.

- Graded Assessment
  - Module 3 Assignment (5 % of total grade)
  - Programming Assignment: Build an Information Extraction Pipeline using Deep Neural Networks (20 % of total grade)

Module 4: Transformers Transform Information Extraction

In this module, you'll dive into the power of deep learning models in diverse fields such as healthcare and sports commentary. You'll learn how to build neural network models that are fine-tuned for specific tasks and discover how to set up a deep neural network for detecting key entities. We'll also introduce you to the world of large language models, showcasing their transformative capabilities and applications in information extraction.

- Graded Assessment
  - Module 4 Assignment (5 % of total grade)
  - Programming Assignment: Jupyter Notebook Practice on Training Transformers (20 % of total grade)

Community Engagement

We expect everyone to be mindful of what they say and its potential impact on others. The goal is to have respectful discussions that do not violate the community space created for these conversations. Here are some productive ways to engage in this course:

- **Participate:** This is a community. Read what others have written and share your thoughts.
- **Stay curious:** Learn from experts and each other by listening and asking questions, not making assumptions.
- **Keep your passion positive:** When replying to a discussion forum post, respond with thoughts on what was said, not about the person who posted. Avoid using all caps, too many exclamation points, or aggressive language.
- **Acknowledge discomfort:** The topics discussed in this course might be challenging or hard to talk about. Stick with it and remember the benefits of having these tough conversations that surface from multiple perspectives.

We expect all learners to abide by our full Learner Engagement Policy. We will specifically be monitoring this course for language that could be considered inflammatory, incivil, racist, or otherwise unacceptable for this learning space, and we will remove language deemed such.

Note regarding "study groups" outside of the platform: While learners are encouraged to interact with one another using communication tools offered within the learning platform, note that any study groups formed outside of this platform, such as over social media or communication applications and websites (e.g., WhatsApp) are not affiliated, endorsed, or moderated by the University of Michigan. If you receive an invitation to an outside study group mentioning the course and claiming to have any official connection to U-M or individual instructors, please exercise caution as this may be an attempted scam.

Academic Honesty

All submitted work should be your own and academic dishonesty is not allowed, such as, but not limited to:

- Copying words, ideas, or other materials from another source without giving credit to the original author
- Copying answers or copying from your peers within the course
- Employing or allowing another person to alter or revise your work, and then submitting the work as your own
- Using Artificial Intelligence tools, such as ChatGPT, to create or edit your work and submitting that work as your own, unless you were instructed to use AI as part of the assignment

Academic dishonesty may result in deleting assignment submissions, forum responses, removal from the course, or other actions as needed. For more information on academic honesty and misconduct, please refer to our Learner Engagement Policy.

Course Support

Questions and discussion of course material should take place within the course itself. Please do not contact instructors or teaching assistants off the platform, as responding to individual questions is virtually impossible.

We encourage you to direct your questions to Course Issues and Questions forum, where your question might be answered by a fellow learner or one of our course team members. For technical help please contact the Coursera Learner Help Center support forums.

Accessibility

We are committed to developing accessible learning experiences for the widest possible audience. We recognize that learners with disabilities (including but not limited to visual impairments, hearing impairments, cognitive disabilities, or motor disabilities) might need more specific accessibility-related support to achieve learning goals in this course.

Please use the accessibility feedback form to let us know about any accessibility challenges such as urgent issues that keep you from making progress in the course (e.g., missing or inadequate alt-text, captioning errors).

Third-party sites and software: While the University of Michigan is not responsible for the accessibility of third-party sites or applications that may be linked from this course, we still encourage you to report associated third-party accessibility issues so that we can ensure you are able to participate. In such cases, we may contact you for additional information as we investigate ways of removing accessibility barriers or to suggest accessible alternatives.

Global Learning Statement

We welcome all learners to this course. By enrolling in this course, you join people from all over the world and we value the perspectives they bring to the learning environment. We strive to create a community of mutual respect and trust where people from all backgrounds and views are valued and heard without the threat of bias, harassment, intimidation, or discrimination. We pay attention to your feedback on how different types of learners experience this course and aim to make improvements so the course can best serve everyone. We hope you enjoy learning about topics that are important to you.

Tools & Technology

Coursera Platform

If you are new to Coursera and you want to familiarize yourself with the platform, read the Help Articles in Coursera Help Center. This is a good resource where you can find answers to many basic questions such as how to adjust video settings, how to submit assignments, and how to earn a course certificate.

H5P

This course includes course content and learning activities built with a tool called H5P to include additional interactivity in this course. These activities will appear within the course itself, to avoid redirecting you outside of the learning platform. If you encounter challenges when exploring the H5P interactive elements, we recommend that you check the course discussion forums on Technical Support for assistance.

GenAI Tools

Please note that any independent use of GenAI tools to complete assignments remains subject to the Academic Honesty section of this syllabus. The use of these tools can also introduce unique copyright, privacy, and ethical considerations. Please review the Center for Academic Innovation's Generative AI Guidance for Learners to learn more.

Coursera Coach

During the course you may encounter Coursera Coach, a generative AI-assisted tool developed by Coursera and supported on the Coursera platform. Using the Coursera Coach is optional. Please note that you should verify AI-generated content for accuracy. The Coursera Coach should not replace your personal analysis of the content.

Jupyter Notebooks

In this course, the Jupyter Notebook tool is used to provide learners a programming environment to work in, directly on the Coursera platform through Coursera Labs. This environment utilizes Jupyter Notebooks to combine code and text.

This environment is intended to provide a ready-made workspace for learners to interact with course materials and without the need for local setup. These notebooks are available for learners to run locally, however support can only be provided for learners who utilize the Coursera Labs environment.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →